

Project Proposal

Pravaha Setu: A System for Coordinating Flood Releases from Multiple Reservoirs *For the Kolhapur and Sangli Region, Upper Krishna Basin, Maharashtra*

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1. Introduction

The Kolhapur and Sangli districts of western Maharashtra have seen repeated and severe flooding in recent years, most seriously during the monsoons of 2019 and 2021. Studies carried out after these events, including the report of the committee set up by the Government of Maharashtra to examine the 2019 Krishna basin floods, have pointed to reservoir operation as one important factor. A large part of the flood peak that reached the two cities was made worse because several upstream dams released water at almost the same time, without a common plan. Koyna, Warna and Radhanagari are each run by a different division of the water resources administration. In the wider basin one major dam, Almatti, lies in Karnataka and is operated by a different state altogether. At present these reservoirs coordinate their releases mainly through telephone calls made during an emergency, and there is no shared system that helps them plan releases together in advance.

The result is that a release which makes sense for one dam on its own can add to the releases from other dams and create a flood peak downstream that is much larger than any single reservoir would have produced. Distance matters here as well. Each dam sits at a different distance from the two cities, so the water it releases takes a different length of time to arrive. These individual flood waves can simultaneously flood the downstream reach when all the dams are opened, potentially causing river overtopping.

This project proposes Pravaha Setu, a system that helps operators to co-ordinate flood releases over several dams. It calculates the anticipated input to each dam using rainfall data, figures out how long

water released from each dam will take to reach downstream cities, and based on those travel times recommends spreading releases over time rather than concentrating them in a single window. The goal is to stop the independent flood peaks from approaching together at Kolhapur and Sangli. The system can also alert if there is space for releasing some water in advance, before a forecast peak arrives, so that reservoir has some spare volume ready. It addresses the rule curve issues highlighted in the official reviews following the floods.

The system only guides. It does not open or close any gate. For every dam, it gives the operator a clear recommendation and a comparison of the likely downstream effects if dams work together versus if they act alone. The operators and the dam authorities make the final call. In its initial phase, the project is limited to three dams (Konya, Warna, Radhanagari) serving the Kolhapur area. Using the real 2019 and 2021 floods, we test the recommendations to estimate how much dam releases increased the peak and how much that could have been avoided.

2. Objectives

- To estimate the water input that is expected at each of the three reservoirs (Konya, Warna and Radhanagari) using rainfall data and past inflow records.
- To work out the travel time and flattening of a flood wave as it moves from each dam to the river stretch near Kolhapur and Sangli, using a standard hydrological routing method.
- To produce coordinated release recommendations that are staggered over time, including advance releases before a forecast peak, so that the combined flood peak downstream is kept as low as possible.
- To present these recommendations to dam operators on an interactive dashboard that compares the suggested coordinated schedule with the current uncoordinated situation, and shows what is likely to happen in each case.
- To test the system against the 2019 and 2021 Krishna basin floods and measure the reduction in the part of the flood peak that comes from dam releases.
- To keep the system advisory at all times, so that operators keep full control over every gate decision.

3. Expected Outcome

Inflow estimates ahead of time. The system will give short term inflow estimates for each reservoir from rainfall and past data. This gives the dam operators an earlier, measured idea of how much water will be heading towards each dam, instead of depending only on what they can see while it happens.

Coordinated releases that account for travel time. By considering the travel time from a dam to the cities, the system will suggest a schedule for the water release that will avoid sending separate flood waves downstream at the same time. It will also suggest advance releases before forecast peak to free up storage.

A dashboard for comparing scenarios. A single detailed dashboard will let operators compare the suggested plan with the current uncoordinated schedule. It will show the expected downriver flow at Kolhapur and Sangli for each case, so the value of coordination is easy to see.

Results backed by real events. Testing the system using 2019 and 2021 flood data will give a clear understanding of how much the dam-related flood peak could have been reduced or neutralized with coordinated operation. This gives the project a concrete result for the report and creates a beneficial impact rather than just a claim on paper.

A design that keeps people in charge. Because the system only works as a helper or advisor, it supports the dam operators and authorities that are already working in the division instead of replacing them. This makes it a realistic tool to use alongside the way departments work today.

4. Literature Review

Forecasting how much water enters and leaves a reservoir is a long standing need in flood management and water planning, and it has become harder as the climate grows less predictable. Physics based hydrological models struggle here because dam operating rules and human decisions are only partly known. This has pushed much recent work towards data driven methods. GarcíaFeal et al. (2022) compared several machine learning methods for predicting reservoir outflow. They found that recurrent networks such as NARX and LSTM performed better than feed forward models, and concluded that memory based models capture the changing behaviour of reservoir operation well enough to support flood warning and dam operation [1].

For predicting inflow from rainfall, Kratzert et al. (2018) showed that Long Short-Term Memory (LSTM) networks can model the rainfall to runoff relationship as well as established hydrological models, and often better [2]. This supports the use of a sequence model for the inflow forecasting part of the proposed system.

Turning a release at a dam into a flood wave that arrives downstream is a classic routing problem. Cunge (1969) examined the Muskingum method in detail and showed it to be a light and widely used way of routing a flood wave through a river reach, along with estimating how long it takes and how much it flattens out on the way [3]. A method of this kind suits a student sized system that needs to estimate the different arrival times of releases from dams at different distances, without going as far as a full hydraulic model.

Operating several reservoirs together has a large research literature of its own. Labadie (2004) reviewed the state of the art in operating multi reservoir systems and set out both the optimisation methods used and the practical obstacles, of which fragmented control among different operating agencies is one of the most common [4]. More directly, Rougé et al. (2021) showed, using real reservoir cascades in the western United States, that modelling or operating each reservoir independently rather than as a coordinated group produces artificial flood and drought errors, and that adding even basic coordination between reservoirs corrects much of this gap [5]. Taken together these two studies make the same point from different angles, that fragmented, independent reservoir operation is a genuine and measurable source of downstream risk, and that coordination is the fix. The proposed system sits in exactly this gap. It does not try to optimise the whole basin at once. Instead it offers a forecast driven advisory layer, aware of travel times, that makes coordinated operation workable for operators who at present rely on calls made during an emergency.

The relevance of this problem to the Krishna basin specifically is documented by Dhumal et al. (2022), who studied the effect of flood releases from the Koyna, Warna and Radhanagari reservoirs during the 2019 Krishna basin floods and examined the role reservoir operation played in the

flooding at Kolhapur and Sangli. These findings emphasize the necessity of an integrated reservoir operation schedule between these dams, instead of the current unconnected reservoir operation[6]. This project answers to that necessity.

5. Scope and Assumptions

To keep the project manageable within a one-year timeframe and set clear expectations, the following boundaries and assumptions are defined.

- Three dams only. The first version covers Koyna, Warna and Radhanagari, which feed the Kolhapur stretch. Almatti in Karnataka and full coordination for the Sangli basin are left for later work, since they bring in coordination between states that is beyond a first prototype.
- Advice, not control. The system gives recommendations and never operates gates. Every release decision stays with the operators and the authorities.
- Hydrological routing, not full hydraulic modelling. The travel time of a flood wave is worked out using a calibrated hydrological routing method, such as the Muskingum method or a calibrated fixed lag, rather than a full two-dimensional hydraulic simulation, which is left for later.
- Only the dam related part of the flood. A large share of the downstream flood comes from local rainfall and small tributaries, for example the local flow into the Panchganga, which no amount of dam coordination can change. For this reason, the results are reported as a reduction in the part of the peak caused by dam releases, and not as prevention of the whole flood.
- Rule based logic in the first version. The first version uses forecast driven rules to stagger releases. More advanced methods, such as reinforcement learning for release planning, are noted as later work.
- Data availability. The project assumes that past inflow, outflow and reservoir level records for the three dams, river gauge readings near Kolhapur, and rainfall data can be obtained from public portals and, where needed, from the departments concerned. When the required figures about an event are sourced from investigation report publications, they are used.

6. Resources Required

To design, build and test the proposed system, the following technical and data resources are needed.

6.1 Software and Development Stack

- **Backend:** A Python service layer built with FastAPI or Flask to hold the forecasting, routing and advisory logic. A Node.js and Express layer can sit in front of it to serve the dashboard through secure REST endpoints.
- **Data analysis and models:** Python 3.9 or later with Pandas, NumPy and scikit-learn for data handling and standard models, and PyTorch or TensorFlow for the sequence models (LSTM) used to forecast input.
- **Hydrological routing:** A custom Python implementation of Muskingum or fixed lag routing, with parameters calibrated on the observed 2019 and 2021 flood records.

- **Frontend:** React.js with Vite and TypeScript, styled with Tailwind CSS and shadcn/ui, to give operators a responsive dashboard.
- **Maps and charts:** Leaflet or Cesium.js for the basin and river map and the dam locations, together with a charting library for inflow, release and downstream flow plots.
- **Database:** PostgreSQL, optionally with the PostGIS extension for spatial data, or MongoDB, for time series and configuration data.
- **Packaging and hosting:** Docker for a reproducible demo, with cloud hosting such as Vercel for the frontend and Render or Railway for the backend for a live demonstration.

6.2 Datasets

- **Reservoir data:** historical inflow, outflow (gate discharge) and reservoir level records for Koyna, Warna and Radhanagari.
- **Downstream flow:** river gauge readings near Kolhapur, such as the Rajaram weir on the Panchganga, along with other relevant Krishna basin gauges.
- **Rainfall:** catchment rainfall data, for example from the India Meteorological Department, for the upstream catchments.
- **Event data:** recorded figures for the 2019 and 2021 floods, for testing.
- **Sources:** India-WRIS, the Central Water Commission, the Maharashtra Water Resources Department, the India Meteorological Department, and published official reports on the floods.

7. Conclusion

Pravaha Setu is built around a real problem: dams upstream of Kolhapur and Sangli release water without talking to each other, and that makes the flooding worse. The system tries to fix this by predicting how much water is coming into each dam, working out how long that water takes to reach the two cities, and using that to suggest releases that don't all land at once. Operators still make every decision, the system only recommends. It covers three dams to start with, and its suggestions are checked against the real 2019 and 2021 floods to see if they'd have actually helped. That keeps the project doable at start, gives a real result to point to, and leaves room to expand later to the full basin, including across state lines.

8. References

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